FMCG Retail Analytics Dashboard: Sales, Customer & Inventory Insights

1. Project Overview

FMCG Retail Analytics Dashboard is a data analytics project developed to analyze retail sales, customer behavior, and inventory performance in the Fast-Moving Consumer Goods (FMCG) sector. The project focuses on converting raw transactional data into meaningful business insights through data preparation, analysis, and visualization.

Python was used for data cleaning, preprocessing, and exploratory data analysis to identify patterns and trends within the dataset. The processed data was then used in Power BI to build an interactive dashboard that provides insights into revenue, profitability, customer segments, sales trends, and inventory metrics.

The dashboard helps in understanding overall business performance and supports data-driven decision-making.

2. Project Objective

The main objective of this project is to develop an interactive FMCG analytics dashboard that helps businesses monitor and evaluate their operational performance.

The project aims to:

- Analyze sales and revenue trends across different categories, cities, channels, and store formats.
- Understand customer demographics and purchasing behavior.
- Evaluate loyalty patterns and their impact on revenue.
- Monitor inventory levels, stock availability, and brand-wise distribution.
- Present key insights through an interactive Power BI dashboard for better decision-making.

3. Data Sources

Raw Dataset : https://raw.githubusercontent.com/Abinaya-1995/Main_Project/refs/heads/main/Indian%20FMCG%20Retail%20Sales%20%20Customer%20%20Inventory%20(2024).csv

Year / Timeline : Data collected during Jan 2024 to Dec 2024

Domain : Retail / E-commerce

Cleaned Dataset : https://raw.githubusercontent.com/Abinaya-1995/Main_Project/refs/heads/main/FMCG_Cleaned_Data_MainProject.csv

4. Problem Statement

FMCG retail businesses generate large amounts of sales, customer, and inventory data, making it difficult to manually track business performance and identify important trends. Without proper analysis, organizations may face challenges in understanding revenue patterns, customer behavior, profitability, and inventory availability.

This project addresses the need for an interactive analytics solution that can transform raw retail data into meaningful insights. By using Python for data analysis and Power BI for visualization, the project aims to provide a clear view of sales performance, customer segments, and inventory metrics to support better business decisions.

5. Dataset Column/features Description

S.NO	Field Name	Data Type	Description
1.	Invoice_ID	Integer	Unique ID assigned to each sales transaction.
2.	Invoice_Date	Date	Date when the sales transaction occurred.
3.	Time	Time	Time of the transaction used for time-based analysis.
4.	Day_Name	Text	Day of the week on which the sale happened.
5.	Month_Name	Text	Month in which the transaction was recorded.
6.	City	Text	Location/city where the transaction took place.
7.	Store_Format	Text	Type of retail store format where the sale occurred.
8.	Category	Text	Product category of the item sold.
9.	Brand	Text	Brand name of the product.
10.	Channel	Text	Sales channel used for the transaction.
11.	Payment_Mode	Text	Payment method used by the customer.

12.	Units	Integer	Quantity of products sold in a transaction.
13.	Cost_Price	Float	Cost price of a single product unit.
14.	Selling_Price	Float	Selling price of a single product unit.
15.	Revenue	Float	Total sales value generated from the transaction.
16.	Cost	Float	Total cost associated with the transaction.
17.	Margin	Float	Profit amount earned after deducting cost from revenue.
18.	Margin_%	Float	Profit percentage calculated from revenue and cost.
19.	Stock_On_Hand	Integer	Available inventory quantity currently in stock.
20.	Reorder_Level	Integer	Minimum stock level at which replenishment is required.
21.	Lead_Time_Days	Integer	Number of days required to receive new stock after ordering.
22.	Customer_Age	Integer	Age of the customer who made the purchase.
23.	Customer_Gender	Text	Gender information of the customer.
24.	Loyalty_Flag	Integer	Customer loyalty status (0 = Non-Loyal, 1 = Loyal).
25.	Profit_Loss	Text	Indicates whether the transaction resulted in profit or loss.

6. Tools & Technologies

Python : Data cleaning, Data transformation, Exploratory Data Analysis, and creating Pivot Tables.

Power BI : Data modelling, DAX calculations, Data Visualization, and interactive Dashboard creation.

Pandas & NumPy : Data manipulation and numerical analysis.

Matplotlib & Seaborn: Data visualization and exploratory analysis.

7. Data Pre-Processing (Python / Power Query)

Tasks Performed:

Initial EDA

```
df
```

	Invoice_ID	Invoice_Date	City	Store_Format	Category	Brand	Channel	Payment_Mode	Units	Cost_Price	...	
0	58018430	02-02-24 13:50	Kolkata	Super	Grocery	Nestle	Online	Wallet	2	148.999035	...	35
1	48157952	09-10-24 11:52	Hyderabad	Hyper	Home Care	PepsiCo	Offline	Wallet	1	80.626759	...	10
2	23283831	26-08-24 22:03	Chennai	Hyper	Snacks	PepsiCo	Online	Card	5	181.233170	...	104
3	53537460	09-06-24 4:34	Bengaluru	Hyper	Beverages	Nestle	Omnichannel	Card	2	195.811403	...	52
4	55348596	07-06-24 1:13	Delhi	Super	Snacks	ITC	Offline	Card	3	23.571888	...	9
...	...	...	...	...	...	...	...	...	...	...	...	
99995	51439182	11-08-24 14:58	Mumbai	Express	Dairy	Britannia	Online	Cash	5	128.336814	...	86
99996	76433622	27-11-24 11:02	Ahmedabad	Express	Dairy	Amul	Offline	Wallet	4	183.100266	...	96
99997	80228351	21-02-24 9:25	Bengaluru	Express	Beverages	ITC	Omnichannel	Wallet	2	163.630389	...	34
99998	49574701	30-03-24 4:11	Chennai	Hyper	Dairy	PepsiCo	Offline	UPI	4	50.230907	...	27
99999	17631041	11-07-24 8:42	Pune	Super	Snacks	ITC	Offline	UPI	2	14.754033	...	3

100000 rows × 21 columns

```
df.head()
```

	Invoice_ID	Invoice_Date	City	Store_Format	Category	Brand	Channel	Payment_Mode	Units	Cost_Price	...	Rever
0	58018430	02-02-24 13:50	Kolkata	Super	Grocery	Nestle	Online	Wallet	2	148.999035	...	350.1144
1	48157952	09-10-24 11:52	Hyderabad	Hyper	Home Care	PepsiCo	Offline	Wallet	1	80.626759	...	106.0341
2	23283831	26-08-24 22:03	Chennai	Hyper	Snacks	PepsiCo	Online	Card	5	181.233170	...	1044.9001
3	53537460	09-06-24 4:34	Bengaluru	Hyper	Beverages	Nestle	Omnichannel	Card	2	195.811403	...	521.7599
4	55348596	07-06-24 1:13	Delhi	Super	Snacks	ITC	Offline	Card	3	23.571888	...	91.8734

5 rows × 21 columns

```
df.columns

Index(['Invoice_ID', 'Invoice_Date', 'City', 'Store_Format', 'Category',
       'Brand', 'Channel', 'Payment_Mode', 'Units', 'Cost_Price',
       'Selling_Price', 'Revenue', 'Cost', 'Margin', 'Margin_%',
       'Stock_On_Hand', 'Reorder_Level', 'Lead_Time_Days', 'Customer_Age',
       'Customer_Gender', 'Loyalty_Flag'],
      dtype='object')
```

```
df.shape

(100000, 21)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 21 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   Invoice_ID        100000 non-null  int64
 1   Invoice_Date      100000 non-null  object
 2   City              100000 non-null  object
 3   Store_Format      100000 non-null  object
 4   Category          100000 non-null  object
 5   Brand             100000 non-null  object
 6   Channel           100000 non-null  object
 7   Payment_Mode      100000 non-null  object
 8   Units             100000 non-null  int64
 9   Cost_Price        100000 non-null  float64
 10  Selling_Price     100000 non-null  float64
 11  Revenue           100000 non-null  float64
 12  Cost              100000 non-null  float64
 13  Margin            100000 non-null  float64
 14  Margin_%          100000 non-null  float64
 15  Stock_On_Hand     100000 non-null  int64
 16  Reorder_Level     100000 non-null  int64
 17  Lead_Time_Days    100000 non-null  int64
 18  Customer_Age      59919 non-null   float64
 19  Customer_Gender   94952 non-null   object
```

```
df.describe()
```

	Invoice_ID	Units	Cost_Price	Selling_Price	Revenue	Cost	Margin	Margin_%	Stock_On_
count	1.000000e+05	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.00
mean	5.512347e+07	2.998290	104.859625	131.107353	393.351350	314.614493	78.736857	0.193368	274.15
std	2.596457e+07	1.414619	54.825341	69.862379	297.154653	234.901995	74.236348	0.075143	129.85
min	1.000120e+07	1.000000	10.000526	10.580915	10.653198	10.002024	0.556870	0.047621	50.00
25%	3.271665e+07	2.000000	57.478508	71.327813	155.415176	125.467072	24.448402	0.131251	161.00
50%	5.518370e+07	3.000000	104.903700	129.999158	313.349100	252.801900	53.552720	0.200287	274.00
75%	7.751346e+07	4.000000	152.320122	188.854907	573.063568	462.376451	110.075103	0.259151	387.00
max	9.999632e+07	5.000000	199.999659	289.699597	1443.134985	999.985219	447.207497	0.310343	499.00

```
df.isnull().sum()
```

	0
Invoice_ID	0
Invoice_Date	0
City	0
Store_Format	0
Category	0
Brand	0
Channel	0
Payment_Mode	0
Units	0
Cost_Price	0
Selling_Price	0
Revenue	0
Cost	0
Margin	0
Margin_%	0

```python
# Checking duplicates
df.duplicated().sum()
```

```
np.int64(0)
```

```python
df.nunique()
```

	0
Invoice_ID	99938
Invoice_Date	91057
City	8
Store_Format	3
Category	8
Brand	8
Channel	3
Payment_Mode	4
Units	5
Cost_Price	99996
Selling_Price	100000
Revenue	100000
Cost	100000
Margin	100000
Margin_%	99979

Data Cleaning & Transformation

- Checked and handled missing values in the dataset.
- Verified data types and corrected inconsistent formats.
- Removed duplicate records to maintain data accuracy.
- Standardized categorical values and text formats.
- Converted date and time columns into suitable formats for analysis.
- Performed data transformation and created required fields for dashboard analysis.
- Prepared the cleaned dataset for Power BI visualization.

Handling Null Values:

```python
# Customer_Age column has 40081 null values
df['Customer_Age'] = df['Customer_Age'].fillna( round(df['Customer_Age'].median()))

# Customer_Gender column has 5048 null values
df['Customer_Gender'] = df['Customer_Gender'].fillna(df['Customer_Gender'].mode()[0])
```

Converting data types:

```
# To convert Invoice_date (Object) data type as 'datetime' format

df['Invoice_Date'] = pd.to_datetime( df['Invoice_Date'] )
```

```
# To convert Customer_Age (Float) data type as 'int' format

df['Customer_Age'] = df['Customer_Age'].astype(int)
```

Splitting column:

```
# To split 'Time' from date column

df.insert(
    df.columns.get_loc('Invoice_Date') + 1,
    'Time',
    df['Invoice_Date'].dt.strftime('%I:%M %p')
)
```

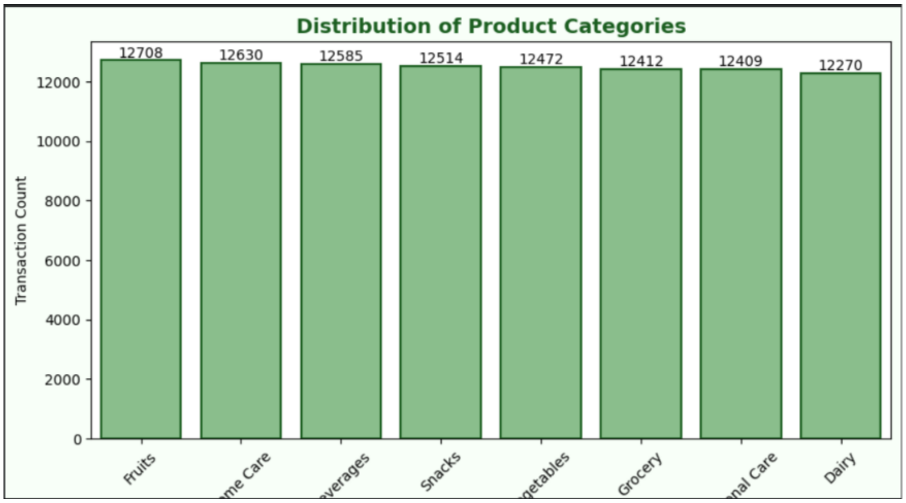
Deriving new fields:

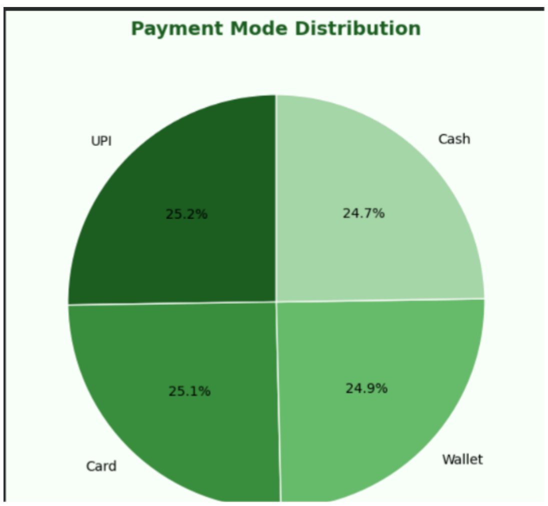
```
# To derive 'day name' from date column

df.insert(
    df.columns.get_loc('Invoice_Date') + 2,
    'Day_Name',
    df['Invoice_Date'].dt.day_name()
)
```

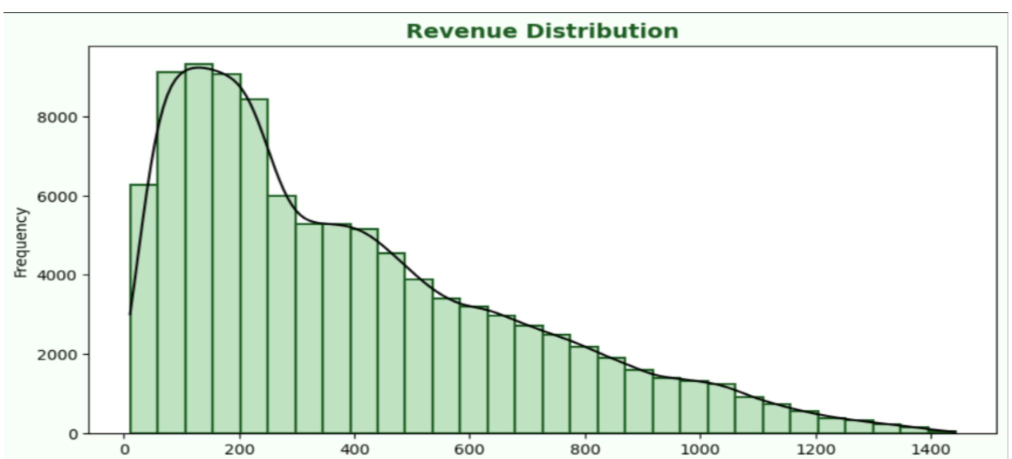
```
# To derive 'month name' from date column

df.insert(
    df.columns.get_loc('Invoice_Date') + 3,
    'Month_Name',
    df['Invoice_Date'].dt.month_name()
)
```

EDA & visualizations

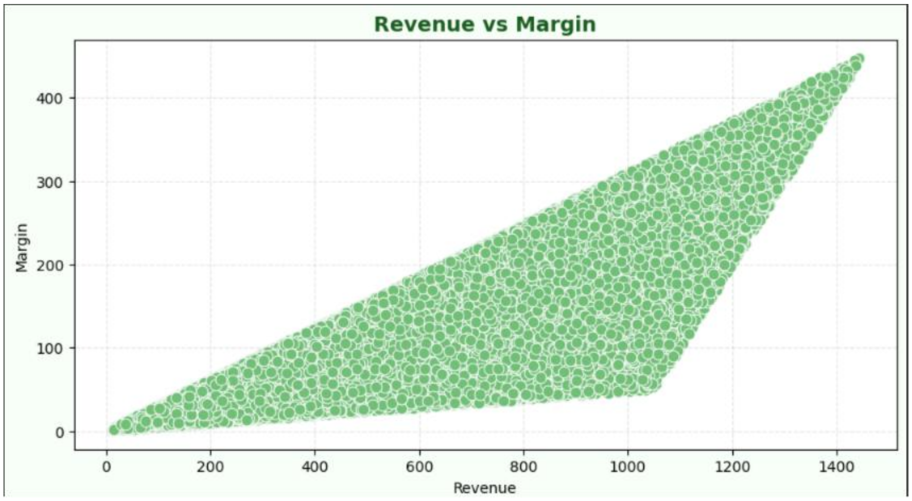
Univariate Analysis:

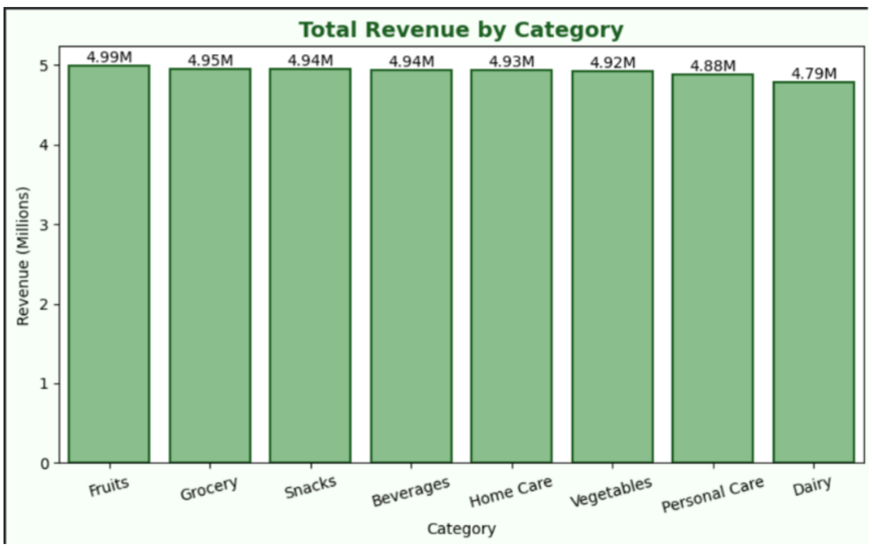




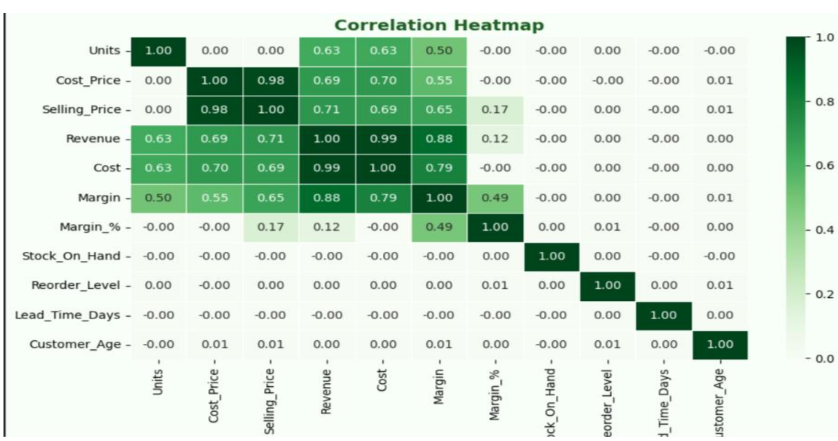


Bivariate Analysis:





Multivariate Analysis:



pivot

City / Category	Ahmedabad	Bengaluru	Chennai	Delhi	Hyderabad	Kolkata	Mumbai	Pune
Beverages	620831.295564	599413.817156	645393.750698	633487.839930	640037.726541	617902.049352	603012.418455	580951.272949
Dairy	576821.284620	592443.464527	625201.290174	622659.697683	585751.752148	580958.469799	598719.448576	605303.882568
Fruits	586208.877734	656489.707270	593713.377618	617954.339026	611523.501138	670316.779136	608818.153447	640393.159999
Grocery	643930.721260	603847.537388	607640.475534	616231.134301	633946.072117	606492.347091	596696.899947	638577.985249
Home Care	618024.831270	600979.432226	622673.877224	618757.780368	619587.034648	622396.379587	617095.523469	611905.616915
Personal Care	623368.446940	610263.071142	590299.675722	627998.785800	620392.059218	607242.269486	611934.008655	592689.130504
Snacks	613580.519016	574516.676935	592491.530183	645713.183405	613674.140702	643278.750837	629367.051538	629932.803593
Vegetables	595008.692872	586276.361762	623245.110419	602459.453404	625874.687783	650878.109336	631956.673301	599602.787434

DAX (Power BI)

Creating measure for Age group:

```
Age_Group = SWITCH( TRUE(), 'FMCG_Cleaned_Data_MainProject'[Customer_Age]  <=25, "18-25", 'FMCG_Cleaned_Data_MainProject'[Customer_Age] <=35,
"26-35", 'FMCG_Cleaned_Data_MainProject'[Customer_Age] <=45, "36-45", 'FMCG_Cleaned_Data_MainProject'[Customer_Age] <=55, "46-55", "56+" )
```

Creating Calculated column for time slot:

```
Time_Slot =
SWITCH(
    TRUE(),
    HOUR(TIMEVALUE('FMCG_Cleaned_Data_MainProject'[Time])) >= 0 && HOUR(TIMEVALUE('FMCG_Cleaned_Data_MainProject'[Time])) < 6, "Night",
    HOUR(TIMEVALUE('FMCG_Cleaned_Data_MainProject'[Time])) >= 6 && HOUR(TIMEVALUE('FMCG_Cleaned_Data_MainProject'[Time])) < 12, "Morning",
    HOUR(TIMEVALUE('FMCG_Cleaned_Data_MainProject'[Time])) >= 12 && HOUR(TIMEVALUE('FMCG_Cleaned_Data_MainProject'[Time])) < 18, "Afternoon",
    "Evening"
)
```

Creating calculated column for Loyalty status:

```
Loyalty_Status = IF( 'FMCG_Cleaned_Data_MainProject'[Loyalty_Flag] =1, "Loyal", "Non-Loyal")
```

Creating measure for avg revenue per customer:

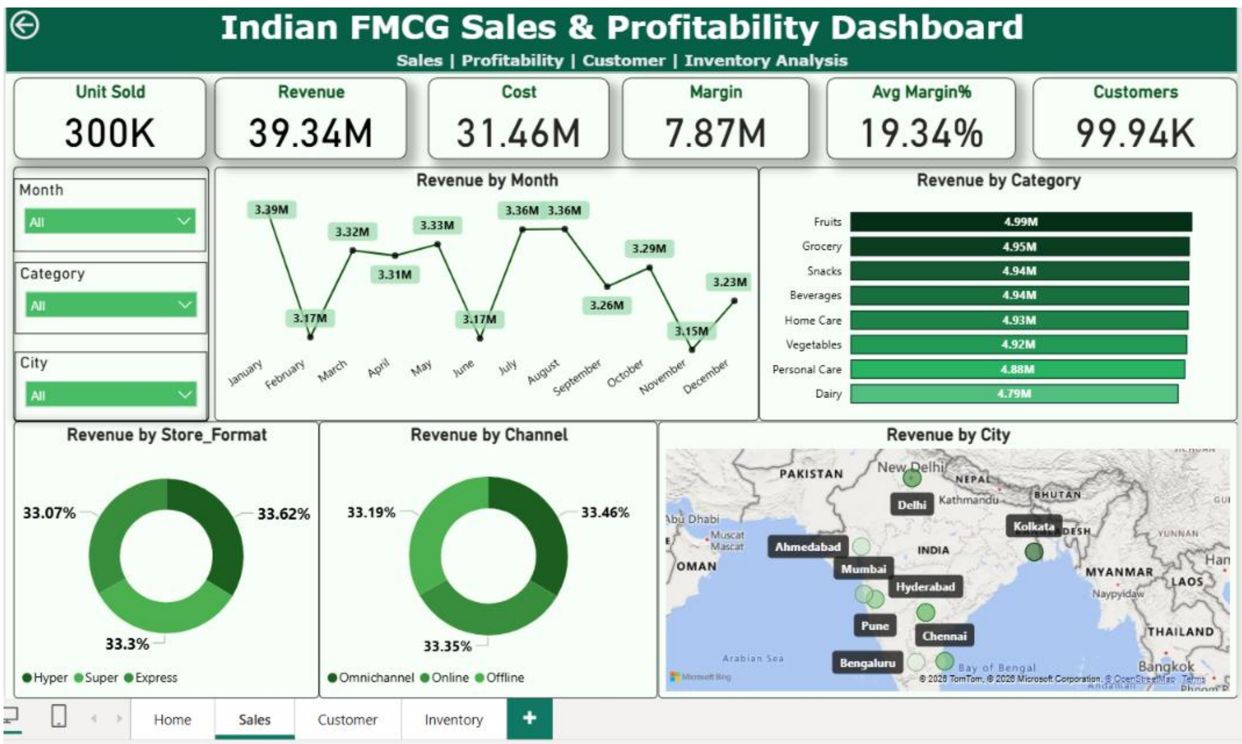
```
1 Avg_Revenue_Per_Customer = DIVIDE( SUM(FMCG_Cleaned_Data_MainProject[Revenue]), DISTINCTCOUNT(FMCG_Cleaned_Data_MainProject[Invoice_ID]) )
```

8. Analysis and Visualizations (Power BI) Dashboard Features:

Multiple Visualizations based on problem statements : Bar charts, Line charts, Pie charts, Cards, Donut chart, Column chart, Scatter chart, Tree map and Matrix to communicate insights.

Dashboard Insights :

Page 1: Sales Analysis Insights



Revenue Performance

- The dashboard provides an overview of overall sales performance using revenue, units sold, and margin metrics.
- Category-wise and city-wise revenue analysis helps identify top-performing business areas.

Category Analysis

- Different product categories contribute differently to total revenue.
- High-performing categories can be prioritized for marketing and inventory planning.

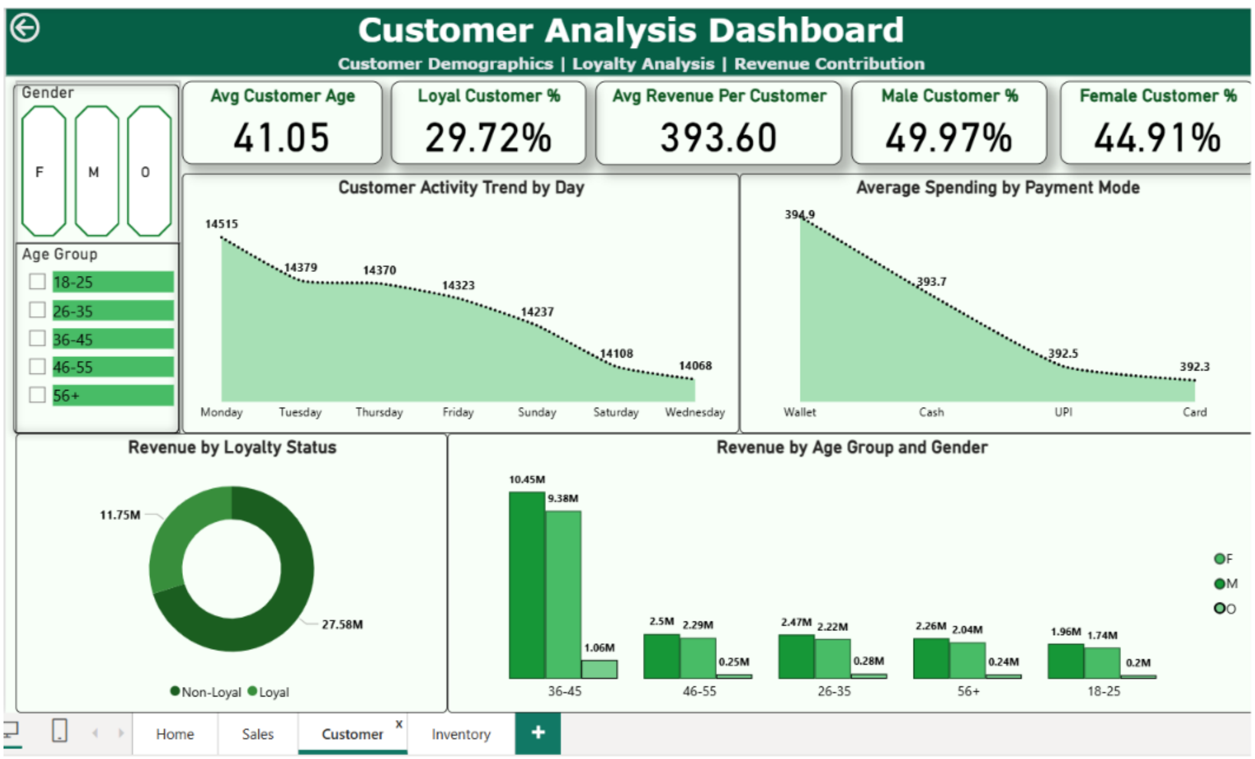
Time-based Analysis

- Monthly and daily sales patterns help identify customer purchase trends.
- Peak sales periods can be used for demand forecasting and promotional planning.

Channel & Payment Analysis

- Sales channels and payment modes provide insights into customer purchasing preferences.
- Businesses can optimize channel strategies based on customer behaviour.

Page 2: Customer Analysis Insights



Customer Activity Analysis

- Customer transactions vary across different days.
- Day-wise analysis helps identify periods of higher customer engagement.

Customer Loyalty Analysis

- Revenue contribution was compared between loyal and non-loyal customers.
- The analysis shows that non-loyal customers contribute higher revenue compared to loyal customers.
- This indicates an opportunity to improve customer retention and loyalty programs.

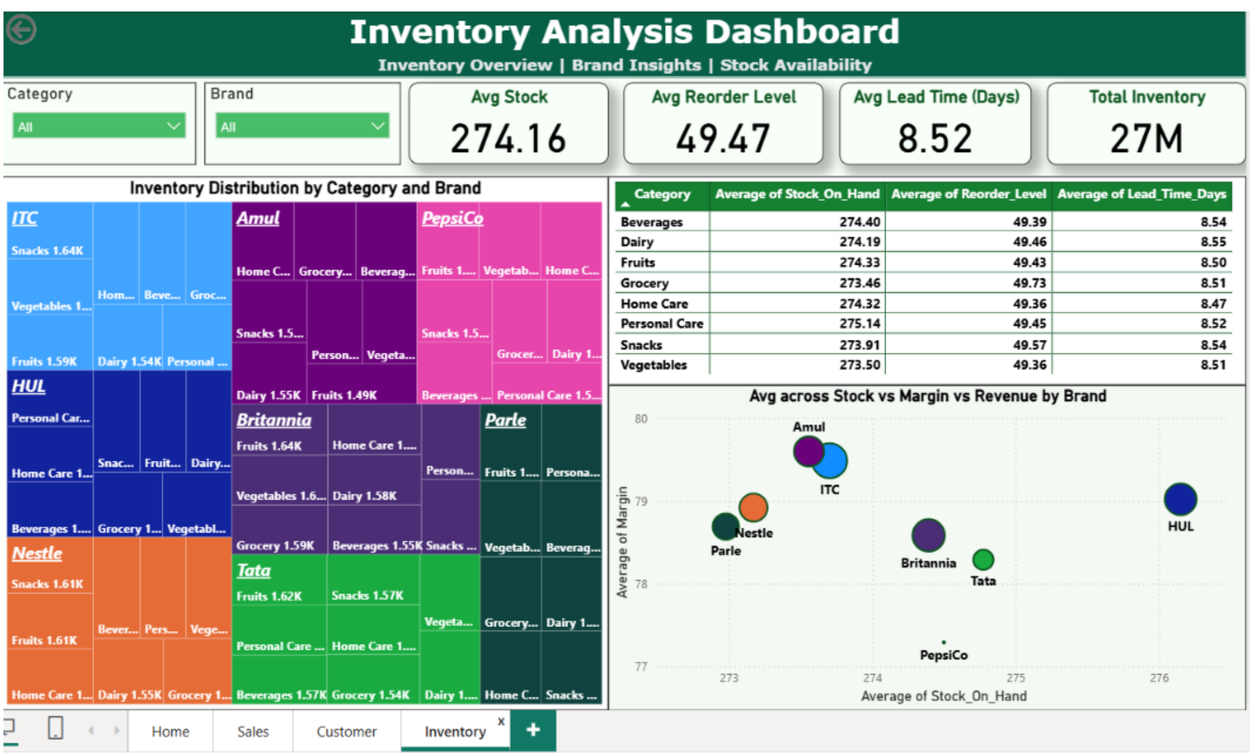
Age Group & Gender Analysis

- Customer segmentation based on age and gender identifies valuable customer groups.
- The **36–45 age group** contributes the highest revenue, indicating this segment is an important customer base.

Payment Behaviour

- Average revenue by payment mode shows customer spending patterns across different payment methods.
- This helps businesses improve payment options and customer convenience.

Page 3: Inventory Analysis Insights



Inventory Distribution

- Stock distribution was analyzed across different categories and brands.
- The tree map identifies brands and categories holding higher inventory levels.

Stock Monitoring

- Stock availability and reorder levels help monitor inventory health.
- Categories with low stock compared to reorder levels require attention to avoid shortages.

Lead Time Analysis

- Lead time analysis helps understand replenishment speed.
- Products with higher lead time require better planning to maintain availability.

Inventory & Profitability

- Inventory performance was compared with profitability metrics.
- Brands with higher stock and higher margins indicate healthy inventory decisions.

9. Data Analysis Insights:

Descriptive Analysis

- The dataset contains 100,000 FMCG retail transactions with sales, customer, product, and inventory details.
- Overall sales performance was analyzed using Revenue, Cost, Units Sold, and Margin.
- Sales data was studied based on category, brand, city, and sales channel to understand business performance.
- Customer details such as age, gender, loyalty status, and payment mode were analyzed to understand customer behaviour.
- Inventory details including stock level, reorder level, and lead time were analyzed to understand inventory status.
- Numerical data was analyzed to identify patterns and trends in the dataset.

Diagnostic Analysis

- Revenue was compared across different customer groups to understand sales patterns.
- Customer analysis showed that the **36–45 age group generates the highest revenue**.
- Loyalty analysis was performed to compare revenue contribution from loyal and non-loyal customers.

- Payment mode analysis helped identify customer purchasing preferences.
- Category and brand analysis helped identify high-performing products.
- Inventory analysis compared **available stock, reorder level, and lead time** to understand stock management.
- Relationships between sales, customer, and inventory features were analyzed to support business decisions.

10. Key Business Findings (Summary)

- The 36 – 45 age group is the highest revenue contributor.
- Non-loyal customers contribute more revenue than loyal customers.
- Customer activity varies across days.
- Payment preferences influence customer spending behaviour.
- Inventory analysis helps identify stock availability and reorder requirements.
- Brand and category analysis helps improve sales and inventory planning.

11. Conclusion

- This project successfully analyzed FMCG retail data using **Python and Power BI** to understand business performance.
- Data cleaning, transformation, and analysis helped convert raw transaction data into meaningful insights.
- The interactive dashboard provides a clear view of sales trends, customer behavior, and inventory performance.
- The insights from this analysis can help businesses improve sales strategies, customer engagement, and inventory planning.
- This project demonstrates the importance of data analytics in making effective business decisions.